INCH-POUND

MIL-P-51433A(EA)
31 August 1988
SUPERSEDING
MIL-P-51433(EA)
5 June 1986

MILITARY SPECIFICATION

PLASTIC TUBE, POLYETHYLENE COPOLYMER

This specification is approved for use by U.S. Army Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of high density plastic tube made of polyethylene copolymer.

1.2 Classification. Plastic tubes shall be of the following types as specified (see 6.1):

Type I - Polyethylene butene-1 copolymer, high density
Type II - Polyethylene hexene-1 copolymer, high density

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.1).

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-SPT-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. :

AMSC N/A FSC 9330

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SPECIFICATIONS

MILITARY

MIL-P-51431 - Plastic Molding and Extrusion Material, Polyethylene Copolymer, High Density

STANDARDS

FEDERAL

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies)

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issue of the documents cited in the solicitation.

ASTM STANDARDS

D 1238 - Flow Rates of Thermoplastics by Extrusion Plastometer
D 1505 - Density of Plastics by the Density-Gradient Technique
D 1599 - Short-Time Hydraulic Failure Pressure of Plastic Pipe, Tubing, and Fittings
D 2122 - Determining Dimensions of Thermoplastic Pipe and Fittings
D 2240 - Rubber Property-Durometer Hardness

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103).

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Material. The plastic tube shall be made from high density polyethylene copolymer conforming to type I or type II of MIL-P-51431 as specified (see 6.1). No additives, coloring material, regrind material, processing lubricants, alcohols, glycols, sugars, organic acids, other compounds containing hydroxyl groups, mercaptans, amines, or other foreign matter shall be added to the virgin polyethylene copolymer or used during the manufacture of the plastic tube. The contractor shall certify that this requirement has been met.

3.2 Infrared spectrum characteristics. The plastic tube shall have the infrared spectrum characteristics specified in 4.4.4.1 when tested as specified therein.

3.3 Water content. The plastic tube shall contain no more than 0.01 percent by weight water when tested as specified in 4.4.4.2.

3.4 Dimensional characteristics.

3.4.1 Length. The length of the plastic tube shall be as specified in the contract or order (see 6.1).

3.4.2 Outside diameter. The plastic tube shall have an outside diameter of 4.730 (+0 or -0.040) inches when measured as roundable tubing.

3.4.3 Wall thickness. The plastic tube shall have an average wall thickness of 0.250 $\pm$ 0.025 inch when measured no less than 1-1/2 inches from an open end. No individual thickness shall be less than 0.200 or greater than 0.300 inch.

3.4.4 Straightness. The straightness of the plastic tube shall be within 0.015 inch per foot when evaluated in lengths no greater than 12 inches.

3.4.5 Out of roundness. The plastic tube shall have an outside diameter no greater than 4.735 inches when measured as non-roundable tubing.

3.5 Physical characteristics. The plastic sheet shall conform to the physical characteristics of table I when tested as specified in 4.4.4.3.

3.6 Workmanship. The plastic tube shall have a smooth finish and shall be free of cracks, blisters, bubbles, discolorations, craze, surface scratches that form indentations, and other defects that could affect appearance or serviceability.

3.7 Preproduction. Prior to the start of regular production, a preproduction sample of plastic tube shall be produced in accordance with this specification for examination and test (see 4.3).

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TABLE I. Physical characteristics

Characteristic	Requirement			
	Type I		Type II	
	Minimum	Maximum	Minimum	Maximum
Burst pressure, pounds per square inch	290*	----	290*	----
Density, grams per cubic centimeter	0.935	0.945	0.937	0.947
Flow rate, grams per 10 minutes	----	2.0	----	2.0
Hardness, type D durometer	60	66	62	68

*The rupture from bursting shall be a hole, tear, or rip showing clear evidence of neck-down of the tube wall around the rupture. If the rupture is a rip, it shall be no longer than 2 inches.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

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4.3 Preproduction inspection.

4.3.1 Lotting. A preproduction lot of 1,000 square feet of plastic tube shall be manufactured using the same methods, materials, equipment, and processes as will be used during regular production.

4.3.2 Sampling.

4.3.2.1 For examination. Twenty-five tubes shall be randomly selected from the preproduction lot.

4.3.2.2 For test. One sample of plastic tube no less than 32 inches long shall be cut from the extrusion line at start up, after every 8 hours of production, after a no production period, on changing lots of polyethylene copolymer, and at the end of the preproduction lot.

4.3.3 Inspection procedure.

4.3.3.1 For examination. The sample tubes selected in 4.3.2.1 shall be examined for the defects listed in 4.4.3.2.

4.3.3.2 For test. The sample shall be tested as specified in 4.4.3.3.

4.3.4 Acceptance/rejection criteria. The preproduction samples shall pass the examinations and tests specified in 4.3.3 to be acceptable. The acceptance number shall be zero. The contractor shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the plastic tube of one type produced by one manufacturer, at one plant, on one extruder, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous.

4.4.2 Sampling.

4.4.2.1 For examination of packaging. Sampling for examination of packaging shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For examination of plastic tube. Sampling for examination of plastic tube shall be conducted in accordance with MIL-STD-105.

4.4.2.3 For physical characteristics tests. One sample of plastic tube no less than 32 inches long shall be taken from the extrusion line at start up, after every 500 feet of production, after a line shutdown and start up, on changing lots of polyethylene copolymer, and at the end of the production.

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4.4.2.4 For infrared spectrum characteristics and water content tests. If more than one lot of tubes is produced from a single lot of resin, one sample of plastic tube no less than 32 inches long shall be taken from the first tube lot and one from the last tube lot produced from the resin lot. If more than one lot of resin is used to produce a single lot of tubes, one sample of plastic tube no less than 32 inches long shall be taken from each resin lot represented. If one lot of resin is used to produce one lot of tubes, one sample shall be taken at the beginning and one sample at the end of the lot.

4.4.3 Inspection procedure.

4.4.3.1 For examination of packaging. The sample unit shall be one filled shipping container. Sample shipping containers and the packaging thereof shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged
(e) Marking incorrect, missing, or illegible

4.4.3.2 For examination of plastic tube. The sample unit shall be one plastic tube having the length specified in the contract or order (see 6.1). Examination for dimensional defects shall be in accordance with ASTM D 2122. Sample tubes shall be examined for the following defects using an AQL of 1.0 defects per hundred units:

(a) Length of tube not as specified
(b) Outside diameter not as specified
(c) Wall thickness not as specified
(d) Straightness not as specified
(e) Out of round
(f) Workmanship (see 3.6)

4.4.3.3 For test. The sample tubing shall be tested as specified in 4.4.4 except that only one sample from the first and one sample from the last lot of tubes produced from each lot of plastic resin shall be tested as specified in 4.4.4.1. Failure of any test by any sample shall be cause for rejection of the lot represented.

4.4.4 Tests. Tests shall be conducted as follows:

4.4.4.1 Infrared spectrum characteristics. Determine infrared spectrum characteristics in accordance with the test procedure for infrared spectrum characteristics in MIL-P-51431.

4.4.4.2 Water content. Using a solids moisture analyzer (see 6.2) and a cylinder of dry nitrogen, regulate the nitrogen pressure to the dryer and the instrument at 5 pounds per square inch (psi). Before installing the electrolytic cell, purge the dryer and the instrument for 10 to 15 minutes. Turn off

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the gas and install the cell, making sure that the cell posts are properly oriented with nozzle fittings inside the cell case. Set the timer in the "Off" position and plug in the power cord. The "Proceeding" lamp will light for approximately 2 minutes and then will shut off automatically. At this time, the "Completed" lamp will light and remain on. With the sample oven sealed and the nitrogen pressure at 5 psi, allow the instrument to dry down for 1/2 hour. Set and adjust the oven temperature to 130°C. Before running a sample, check the electrical zero of the integrator by pressing the "Zero" push button and observing the integrator count rate. The "Zero Adjust" control should be set so that the count rate is about 2 to 3 micrograms per minute. Insert the sample boat to be used in the instrument oven and allow it to dry for 30 minutes. Remove the empty boat from the oven for 10 to 15 seconds and then replace it in the oven and record the count for 30 minutes. Zero the integrator. Working as quickly as possible to avoid gaining or losing moisture, transfer approximately 0.5 gram of sample from a previously weighed weighing bottle to the empty sample boat which has been kept dry, remove the oven cap, insert the sample boat into the heated area using tweezers, replace the oven cap and turn to lock, and then turn the timer on. Set the timer for the time limit desired. For polyethylene, 30 minutes is usually sufficient time to drive off all the water. For the first sample, set the timer for maximum time in order to establish the proper running time needed to drive out all the water. Allow the instrument to run until the integrator runs very slowly. Then turn the timer off and record the count. Once the proper running time is established, the timer can be set to automatically turn the oven and the integrator off and to turn the oven blower on. The blower will continue to run until the oven temperature has been lowered to ambient. A new sample should not be placed in the instrument until the "Completed" lamp lights. Calculate the percent by weight water as follows:

$$\text{Percent water} = \frac{(A - B)(10^{-4})}{W}$$

where: A = Microgram count for sample,
B = Microgram count for empty sample boat, and
C = Weight of sample in grams.

4.4.4.3 Physical characteristics. Determine the physical characteristics of the sample specified in table I in accordance with the test methods specified in table II.

TABLE II. Test methods for physical characteristics

Characteristic	ASTM test method
Burst pressure	D 1599
Density	D 1505
Flow rate	D 1238, procedure A, condition F
Hardness, type D durometer	D 2240

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5. PACKAGING

5.1 Interplant shipment (see 6.3). A quantity of plastic tubes as specified in the contract or order (see 6.1) shall be unit packed and packed in a manner to provide adequate protection from physical damage or deformation and exposure to light from the supply source to the first receiving activity for immediate use or further processing. Tubes shall be packed in a vertical configuration and fully protected against damage which would affect the roundness of the tubes during shipment and storage. Shipping containers shall be in compliance with the rules and regulations applicable to the mode of transportation.

5.2 Marking. Shipping containers shall be marked in accordance with Fed. Std. No. 123 and shall include the contractor's name, address, contract number, and date of manufacture.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification
(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2)
(c) Type of material required (see 1.2 and 3.1)
(d) Length of plastic tube required (see 3.4.1 and 4.4.3.2)
(e) Quantity of tubes per shipping container (see 5.1)
(f) Preproduction:

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for Government to notify the contractor whether or not to proceed with production.

6.2 Solids moisture analyzer. A DuPont type 26-321A moisture analyzer has been found satisfactory for use in the water content test.

6.3 Interplant shipment. Unit packing and packing for interplant shipment is designated for supplies and materials that do not enter the military supply system. Typical interplant shipments are those from a vendor to a subcontractor or a prime contractor, or from a subcontractor to a prime contractor, or from a vendor to a contractor to a military arsenal or plant.

6.4 Extrusion of plastic tube. One combination of equipment and operating conditions that may be used for extruding plastic tube is described in the appendix to this specification.

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6.5 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E 29.

6.6 Subject term (key word) listing.

Polyethylene butene-1 copolymer
Polyethylene hexene-1 copolymer

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 9330-A013

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APPENDIX

EXTRUSION OF PLASTIC SHEET

10. SCOPE

10.1 This appendix describes equipment and operating conditions that have been used in the extrusion of plastic tube. This information is furnished with the distinct understandings:

(a) That the contractor is not required to follow the method.

(b) That the requirements of this specification must be met regardless of the method of manufacture.

(c) That the Government makes no warranty that plastic tube made with equipment described and under the conditions described will meet the requirements of this specification.

20. EQUIPMENT

20.1 Extruder. Use an extruder that is 2-1/2 inches in diameter and has a 24 to 1 L/D screw and a 40-horsepower drive motor.

20.2 Screw. Use a screw with nine turns feed 0.375 inch deep, nine turns transition - constant taper, six turns meter 0.150 inch deep, and a pitch of 2-1/2 inches.

20.3 Die. Use a die with a 5.438-inch bushing and a 4.800-inch mandrel.

20.4 Vacuum sizer. Use a 6-foot Conair vacuum sizer or equivalent.

20.5 Cooling tanks. The cooling tanks should be approximately 20 feet.

20.6 Puller. Use a Gatto Model No. 1153 puller or equivalent.

20.7 Saw. Use an Emery Model No. 4 saw or equivalent.

30. OPERATING CONDITIONS

30.1 The extruder temperature profile should be as listed in table III.

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TABLE III. Extruder temperature profile

Characteristic	Condition
Zone 1 temperature	480°F
Zone 2 temperature	480°F
Zone 3 temperature	460°F
Zone 4 temperature	465°F
Die 1 temperature	470°F
Die 2 temperature	465°F
Die 3 temperature	520°F
Stock temperature	500°F
Water temperature	50°F
Motor load, amps	45
Line speed, feet per minute	1.25
Screw rpm	64
Head pressure, psi	2600

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-51433A(EA)	PLASTIC TUBE, POLYETHYLENE COPOLYMER

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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